

The picture can't be displayed.

Master Degree in *Physics of Data*

Prof. Marco Zanetti

FORMER! President of Physics of Data faculty council

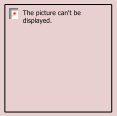
Key to access the moodle page of Laboratory of Computational Physics mod A:

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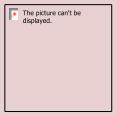
Relevant things



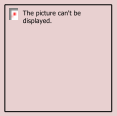
What Physics of Data is about



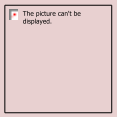
Are you in the right place?



If yes, you are going to profit greatly from the program!

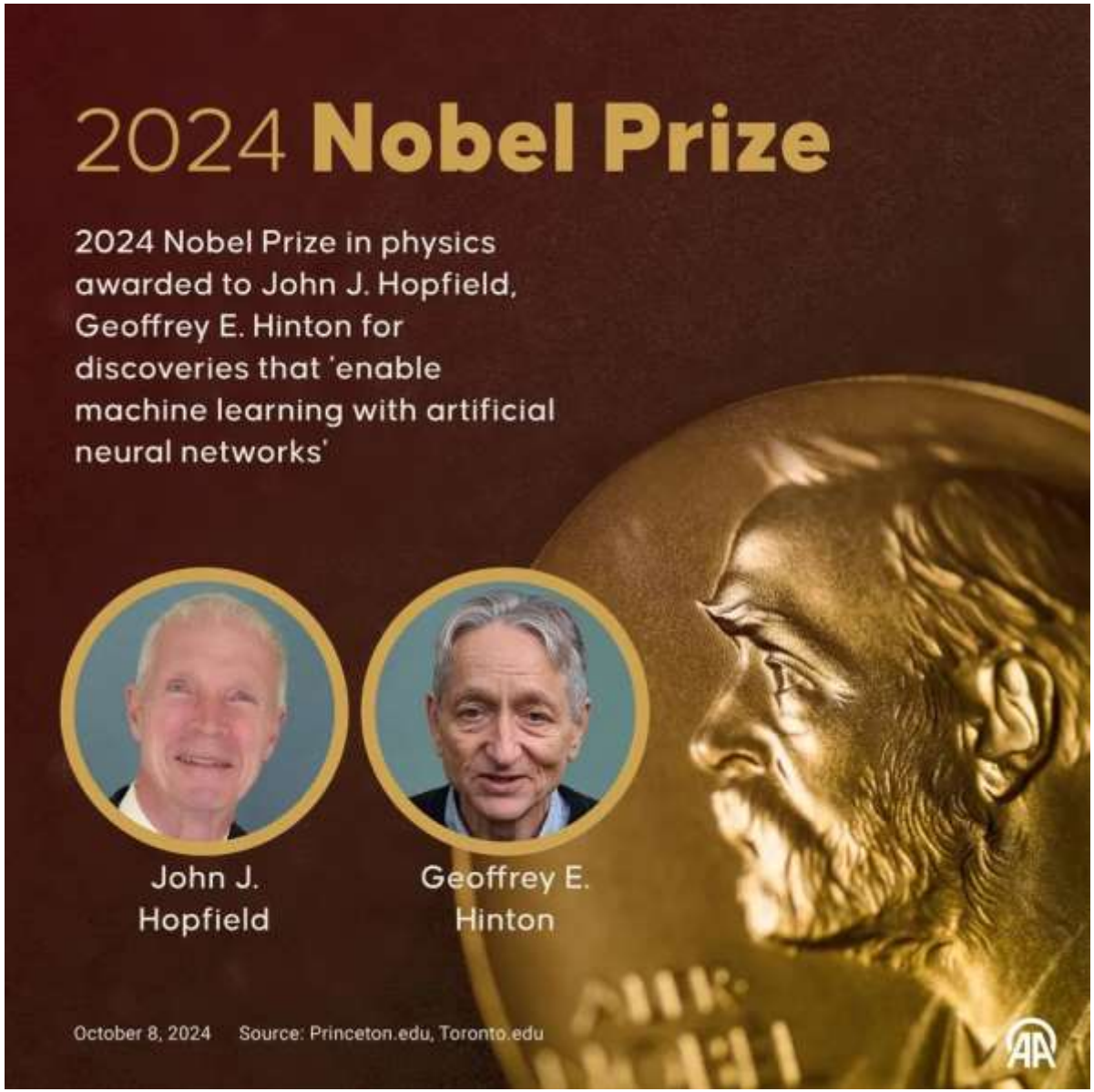


Study plan: tracks, mandatory/optional courses and all that



Attitude/spirit, wherabouts, bureaucracy

A visionary strategy



The project in a nutshell

- A Master Degree aiming at combine the classic competences of a Physicist with those of a Data Scientist
  - "Classe di Laurea LM17", i.e. we are graduating physicists!
- Gathering, processing, interpreting complex and large datasets is the core of modern scientific research as well as the base of the digital revolution
  - At the same time, possible to specialize in any Physics domain
- Twofold training goal
  - Provide physicists with tools to effectively tackle modern science challenges
  - Match the job market requests
- "Learning by doing" is the guiding principle ➔ internship!

The name

- "Physics of Data"
  - From a seminal Nature commentary that elaborates on how connected Physics and Data Science are

commentary

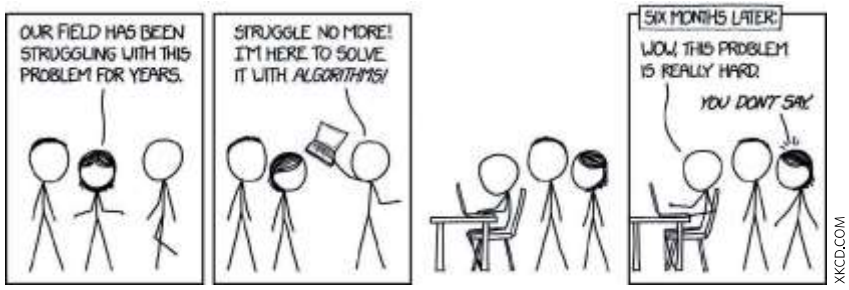
The physics of data

Jeff Byers

Physicists are accustomed to dealing with large datasets, yet they are fortunate in that the quality of their experimental data is very good. The onset of big data has led to an explosion of datasets with a far more complex structure — a development that requires new tools and a different mindset.

Ernest Rutherford is said to have famously declared that "If you need statistics then you should have done a better experiment." This observation still strikes a chord with most physicists today: complex, messy experiments tend to be viewed as bad experiments, and poor data quality is, often correctly, attributed to poor experimental design.

Then again, I guess Rutherford would have made a terrible astronomer. Astronomers have to follow a vastly different path than physicists, since in their



Learning by doing

- The key aspect of the program consists of the hands-on activities and research projects
  - You don't learn only by taking notes but especially doing stuff yourself
- Several classes envisage lab activities. For most of the classes the final exam is based on a research project
  - Projects are carried out by groups of students with the supervision of experts (professors and collaborators)
  - Those project will be items of paramount importance in your CV!!
- In addition to theoretical aspects, we want students to develop practical skills, especially those related to modern technologies

Generalities

- All classes are in English
- Mandatory internship in a private company or research institute/foreign university
- Close collaboration with related masters (in particular with *"Physics", "Astrophysics and Cosmology" , "ICT for internet and multimedia", "Data Science"*)
- High-profile visiting professors from all over the world
  - Skilled teaching assistants from companies and research institutes

The graduate's profile

- The educational path can be tuned with great flexibility! Students are totally free to focus on whatever subject in Physics!
  - Fundamental interactions, Statistical Mechanics, Material Science Astrophysics/cosmology, Medical/Bio Physics, Climate Physics, etc.

- The common ground to all students is the expertise in data analysis, advanced statistics, machine learning, complex systems, data management and processing, modern software (python/ big data analytics) and hardware tools (FPGA, GPU), quantum computing
  - In-house programs and classes from Engineering degrees (ICT)

- "Tracks" are provided as examples of study plans (see later)

The graduate's profile

- All this is precisely what is needed for modern research in physics domains involving data

- All this will allow you to cross boundaries among scientific disciplines
  - Build models from data, whatever data are dealt with!
  - Interdisciplinarity is the key for future scientific research

- All this includes and extends what is required to a Data Scientist by the job market
  - And you'll have a (public) portfolio to base your CV on!

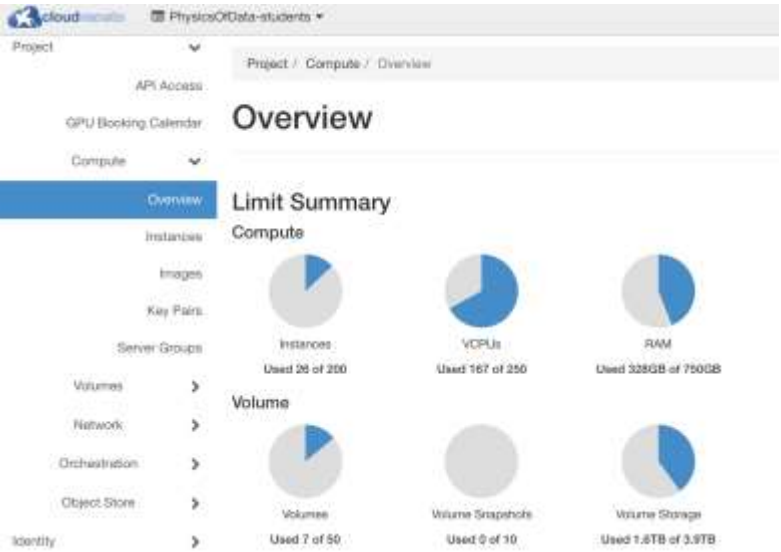
Labs and resources





- Students can profit from excellent laboratories, with modern hardware and software technologies
  - Computing lab with >100 desktops
  - Electronics lab

Labs and resources



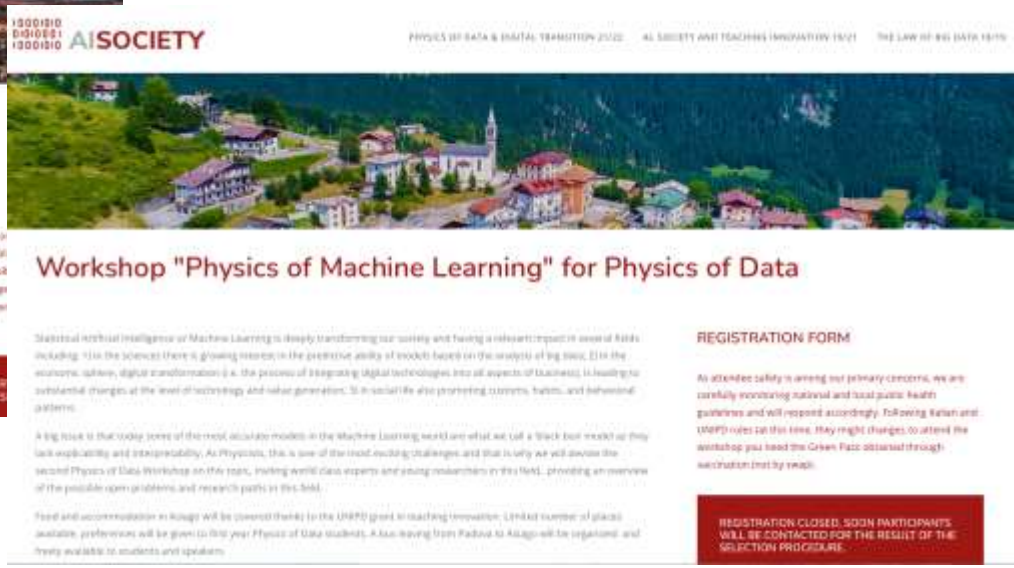


- Our own private Cloud infrastructure (Cloud Veneto)
  - 200 Virtual Machines, 250 VCPU, 750 GB of memory, 4 TB of disks
  - 4 GPU (Nvidia T4)
- A variety of dedicated hardware, available to students:
  - FPGA boards (Artix), NUC, PCB and microcontrollers, Nvidia Jetson, etc.
- 4 Smart-boards

Events

AI Society





- We organize conferences for students, where keynote speakers and alumni give seminars on several topics
- Two events per year, with expenses for students completely covered by Univerisity
- A unique opportunity for going deep on modern applications and networking with researchers, professionals and peers

Events









Events







Internship

- Out of 4 semesters, the last one is entirely dedicated to an internship
  - The work done during the internship typically constitutes the subject of the final thesis (even though a separate thesis work can be done)
- Internships can be done either at research centres or at companies
  - Not necessarily defining your next career's path, it could be taken as "let me have a look" experience
  - Anyhow, a great opportunity to get boost your career!
- A great success so far!

Internship at research centres

- Padova Neuroscience Center (Padova): Computational Neurosciance, Imaging, Data Analysis.
  - Corbetta / Suweis
- CNISM (Padova): Physics of Matter, Biological Physics
  - Seno, Maritan
- CNR (Roma) : Network applied to Economy/Finance and Neuroscience
  - Gabrielli / Baiesi, Suweis
- CNR (Padova): Plasma physics, network controllability and modelling
  - Sattin / Baiesi, Suweis
- CNRS (Paris-Saclay): Disordered systems, Computational Physics
  - Rosso/ Baiesi

- FBK (Trento): Complex System, Machine Learning
  - De Domenico / Suweis
- FBK (Trento): Biomedicin, Envirnoment, High Energy Physics, Machine Learning
  - Cristoforetti / Zanetti
- FBK (Trento): IoT, Big Data, Data Science
  - Salvadori / Zanetti
- ARPAV (Rovigo): Environmental physics, statistics, machine learning
  - Menini / Garfagnini
- Azienda Ospedaliera (Padova): Machine Learning, Imaging
  - Cipriani (Cardiology) / Zanetti, Suweis
  - ? (Patological anatomy) / Suweis

- ENS (Paris): Disordered systems, Computational Physics
  - Ros / Baiesi
- IFISC (Palma de Majorca): Complex Systems
  - Meloni / Suweis
- IMT (Lucca): Network applied to Economy/Finance and Social Networks
  - Caldarelli / Suweis
- ICTP (Trieste): Ecological modelling
  - Grilli / Maritan
- HiT (Padova): Computer Vision, Machine Learning
  - Ballan / Suweis
- CiG (Bologna): Metagenomics, Machine Learning
  - Castellani / Suweis

- JPARK (Tokai): Neutrino Physics
  - Takahashi / Collazuol
- INFN (Padova): High energy and Neutrino Physics, Imaging, Machine Learning
  - Zanetti, Collazuol, Garfagnigni
- CERN (Geneva): High energy physics, big data, machine learning, computing infrastructure
  - Campana (Computing infrastructure) / Zanetti
  - Meschi (Data Acquisition systems) / Zanetti
  - Canali (Big Data analytics) / Zanetti
  - Roncarolo (Beam diagnostic) / Zanetti

And many more!!

Internships at companies

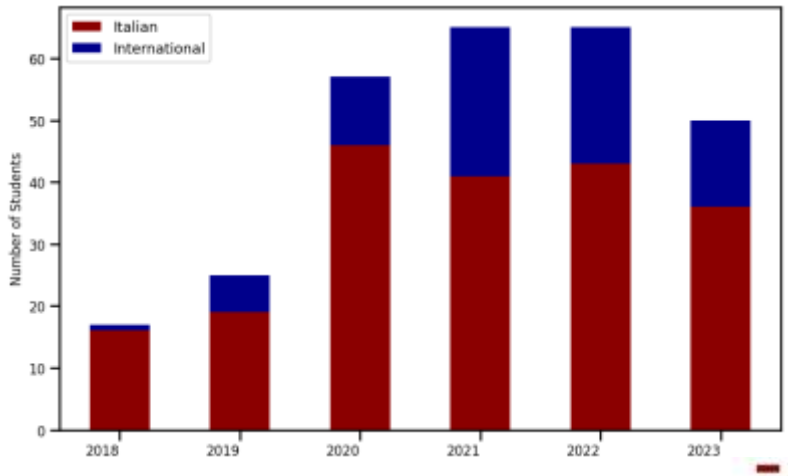


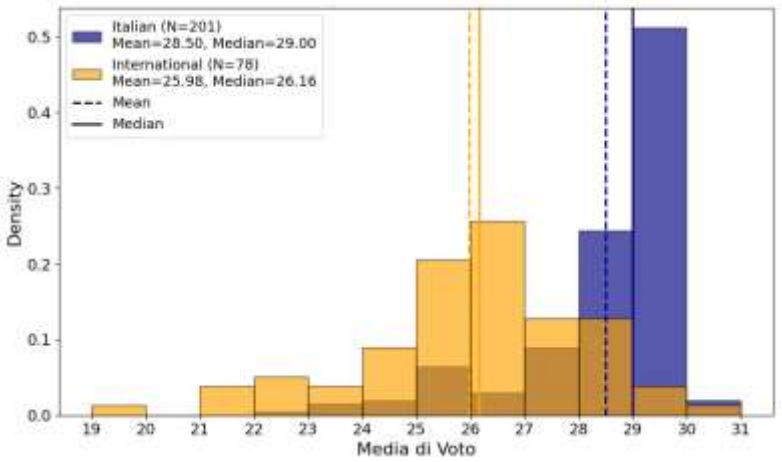
And many more!!

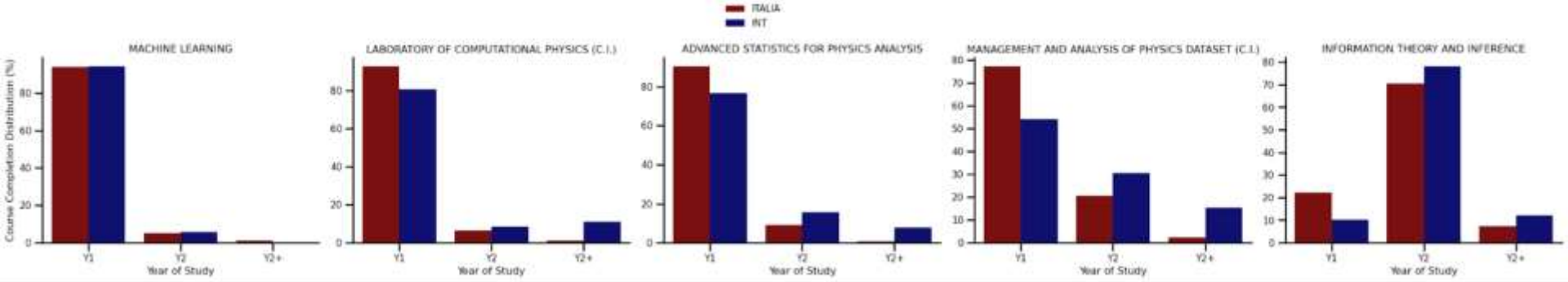
UBS
Akronos technologies
PORINI A DGS COMPANY
EXOR INTERNATIONAL
pwc
ENERYIELD
OMNYS
XOR Capital
Radarmeteo
Bending Spoons
Finapp "Life from Cosmos"
E4 ANALYTICS
FATER:R Growing for a better lifer life
NOZOMI NETWORKS
221e

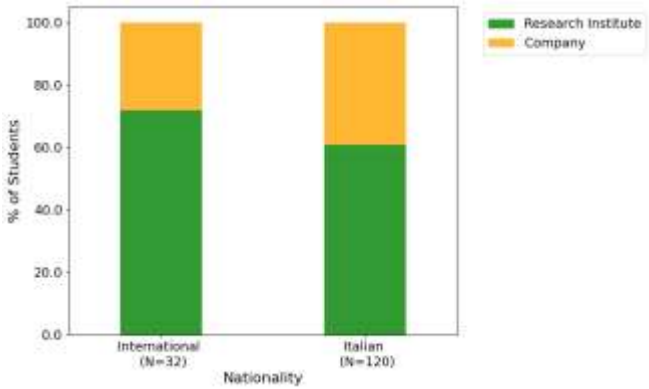
More on the document listing our graduates so far

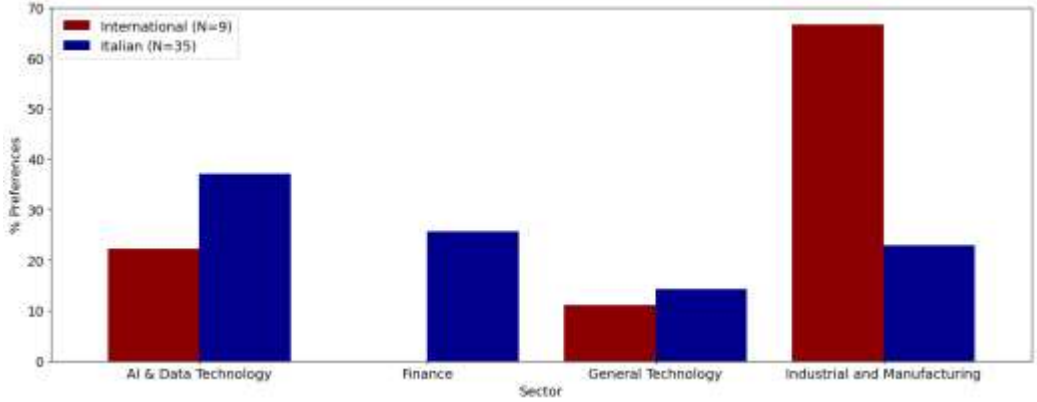
Some stats


Italian
International
Number of Students
2018
2019
2020
2021
2022
2023


Italian (N=201)
Mean=28.50, Median=29.00
International (N=78)
Mean=25.98, Median=26.16
Mean
Median
Density
Media di Voto


ITALIA
INT
MACHINE LEARNING
LABORATORY OF COMPUTATIONAL PHYSICS (C.I.)
ADVANCED STATISTICS FOR PHYSICS ANALYSIS
MANAGEMENT AND ANALYSIS OF PHYSICS DATASET (C.I.)
INFORMATION THEORY AND INFERENCE
Course Completion Distribution (%)
Year of Study
Y1
Y2
Y2+


Research Institute
Company
% of Students
International (N=32)
Italian (N=120)
Nationality


International (N=9)
Italian (N=35)
% Preferences
AI & Data Technology
Finance
General Technology
Industrial and Manufacturing
Sector

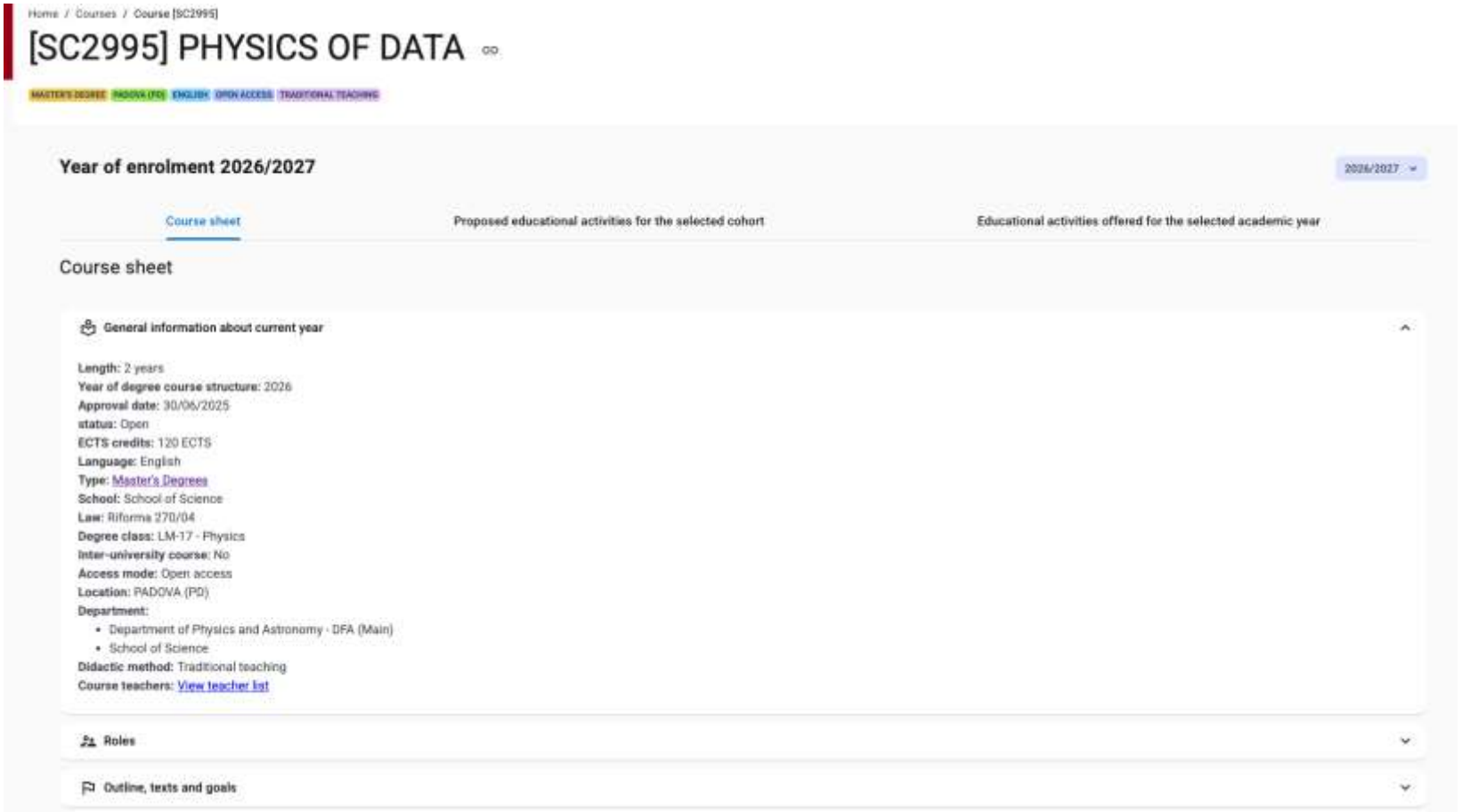
PoD web site



[Check it out!]

Course Catalogue (syllabi)

- Available at this link
- Check for information related to each course in the didactical offer



Study plan

- At some point students need to compile a study plan
  - There will be 2, 1-month long, time-windows for such compilation
  - Exceptional submissions and approval are anyway possible at any time
- That is not such concept of “enrolling” to a course
  - You register (respecting the corresponding schedules) to individual exams

A word of caution

- If you DON’T have a scholarship, they study plan matters for administrative purposes
- If you DO have a scholarship, they study plan is a necessary condition to get the scholarship
- If you plan and pass the mandatory exams, you are on the good path

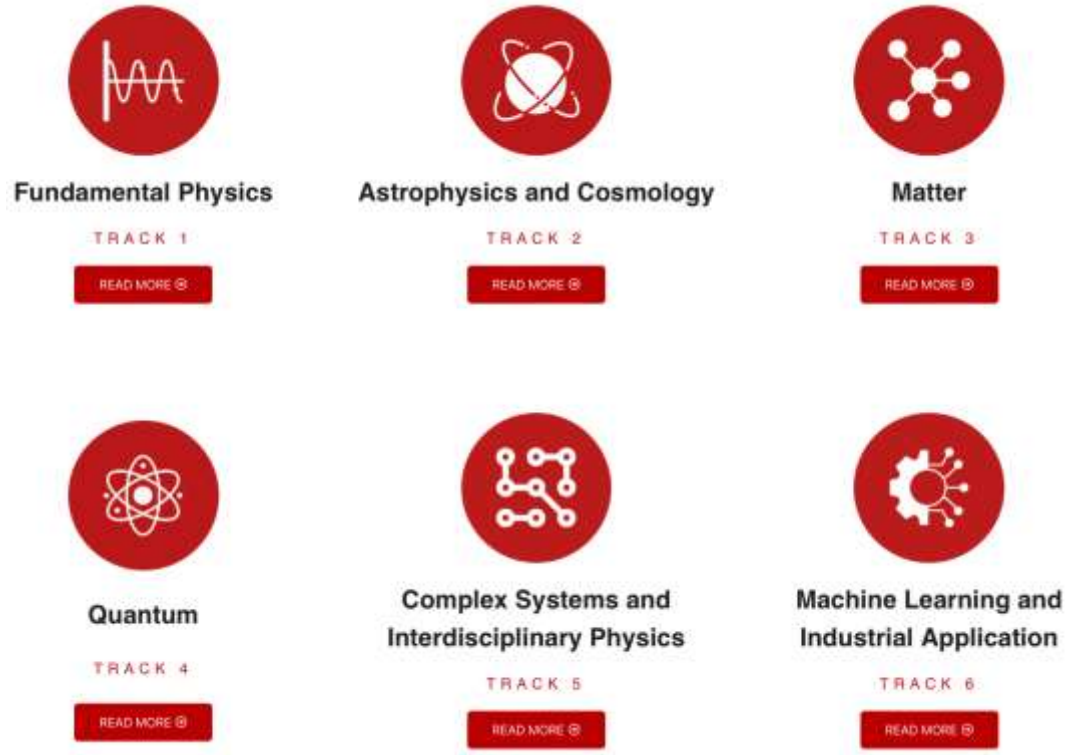
Study plan

- Each course has a code (SSD) associated to (e.g. FIS/01): that represents the area/discipline
- Each student has to gather enough credits (CFU) for specific SSDs in order to graduate
- Taking all the mandatory courses garantees the minium criteria to graduate
- The study plan needs be approved by a dedicated panel of Physcis of Data faculty members
  - To check whether the conditions for graduation are met
  - To check whether it makes sense from a didactical point of view
- Still basic, obvious rules apply
  - You cannot take a “simpler” version of one of our exams
  - N.B. you need to have your plan approved before attempting an exam!

Study plan

- If you have doubts, contact the professors in charge of approving the study plans:
  - Azaele, Pazzini, Triossi, Liguori, Suweis
- N.B.: a full study plan can be properly compiled only during the second year. This means that during the first year, what matters is the list of the exams of the first year

Study plan



Newly added:
TRACK 7, Climate Physics

- There is quite some flexibility in chosing exams beyond the mandatory ones.
- In order to assist students, we provided some EXAMPLES of study plan
- We grouped them in “tracks” associated with possible areas of interest
  - N.B.: tracks have no official meaning, they are not curricula!
- Students are invited to browse the web site for more information about the tracks
- Dedicated meetings during the year: next is

Courses, 1st semester

- 5 “6 CFU” courses
  - Laboratory of Computational Physics, mod A (lab)
  - Management and Analysis of Physics Datasets, mod A (lab)
    - Completely refurbished w.r.t past years
  - Machine Learning
  - A “theory” course ← choose between 3
  - A “physics” course (e.g. of type FIS-0*) ← choose between many
- That’s a lot of stuff.. Take it easy! I.e. gauge your effort to a subset of them
  - You can try and attend all of them, in particular the two labs

Labs

- Laboratory of Computational Physics
  - Mod A main goal is to bring everybody to level, i.e. filling the gaps
    - For many it's going to be easy, for others not so much..
  - Mod B is split into two
    - First 24h: applications of machine learning techniques to Physics, common to everyone
    - Second 24h: a lab of physics data analysis, students split into groups
- Management and Analysis of Physics Datasets
  - Mod A, the "hardware" course
    - Build a computer from scratch!
  - Mod B, the "big-data", i.e. data processing and data management
  - (Both are followed by optional advanced courses at the second year)

Theory Courses

- 3 options, to match the 3 areas (fundamental interactions, statistical mechanics, astrocosmo)
- Theoretical Physics ➔ Fundamental Interactions
  - It's actually the first module of a 12 CFU course. Several students of that area take both of them (the second module counting as the "physics course")
  - N.B. the first 6 CFU are thought in the first half of the semester
  - N.B. there is a synthetized version of the 12 CFU course, *Introduction to Field Theory*, offered by Astrophysics and Cosmology, but at the second semester
- Models of Theoretical Physics ➔ Statistical mechanics
  - Little overlap with the actual "Statistical mechanics" course, i.e. you can take both of them
- General Relativity ➔ Astrophysics and Cosmology

The "physics course"

- As 5th course you need to take a "physics" one, typically chosen from the offer in the area you'd like to specialise
- We provide a list of optional courses, but in reality, you can pick whatever you want.
  - Our choices are based on the characterizing courses from "Physics" and "Astrophysics and Cosmology" master programs
- Examples:
  - A physics lab
  - The whole Theoretical Physics
  - Fundamentals of Astrophysics and Cosmology
  - Quantum Information with atoms and photons
  - …

References

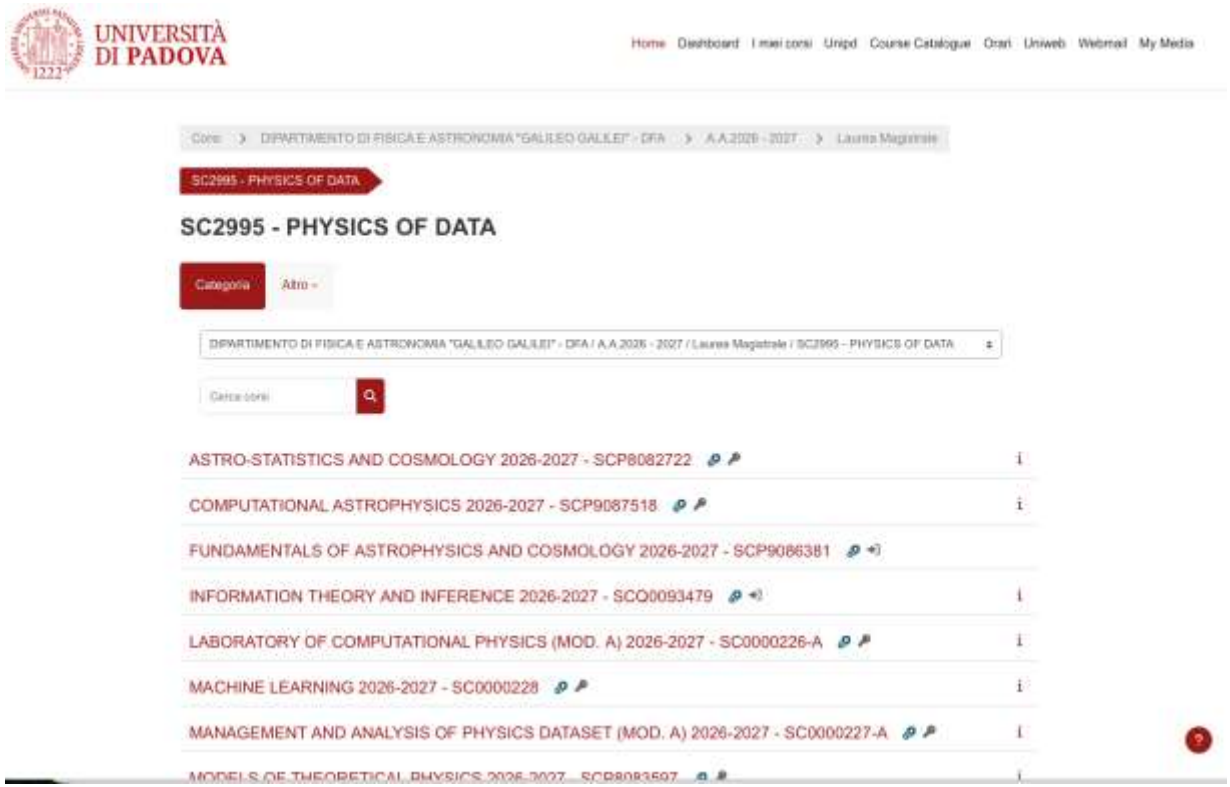
- Fundamental Physics
  - Marco Zanetti, Jacopo Pazzini, Andrea Triossi, Alberto Garfagnini
- Statistical Mechanics, Complex Systems, Biophysics/Neuroscience
  - Samir Suweis, Sandro Azaele, Monika Fuxreiter, Chiara Poletto, Manlio Di Domenico, Gianmaria Falasco
- Quantum Information
  - Simone Montangero, Ilaria Siloi, Marco Di Liberto, Pietro Silvi
- Astrophysics and Cosmology
  - Tiziano Zingales, Alessandro Renzi, (Michela Mapelli from Heidelberg)
- Industrial applications
  - Marco Zanetti, Samir Suweis, Manlio Di Domenico, Jacopo Pazzini, Alberto Garfagnini

Notes for foreign students

- Things are very likely different at our University then what you are used to. Cultural differences might affect your accademic and social experience
- The level of the courses/program shall be different with what you are used to. That might well be easier than you expect, but in some cases is harder
- Keep in mind that there are rules that need to be respected, both from we faculty and by you students
  - E.g. exams cannot be taken and registered at any time in the year. There are well defined exam sessions (Jan-Feb, Jun-July, Sept)
- Rules might sometimes be non-intuitive, please be sure you are aware of them, consult with secretariats for doubts:
  - didattica.dfa@unipd.it scienze.studenti@unipd.it global.engagement@unipd.it
  - Consult with professors ONLY for matters related to didactics or research!
- Constrains requested to renew scholarships must not be used as leverage to violates rules

Moodle

- Available at this link
- Please subscribe to the courses you'd like to attend!



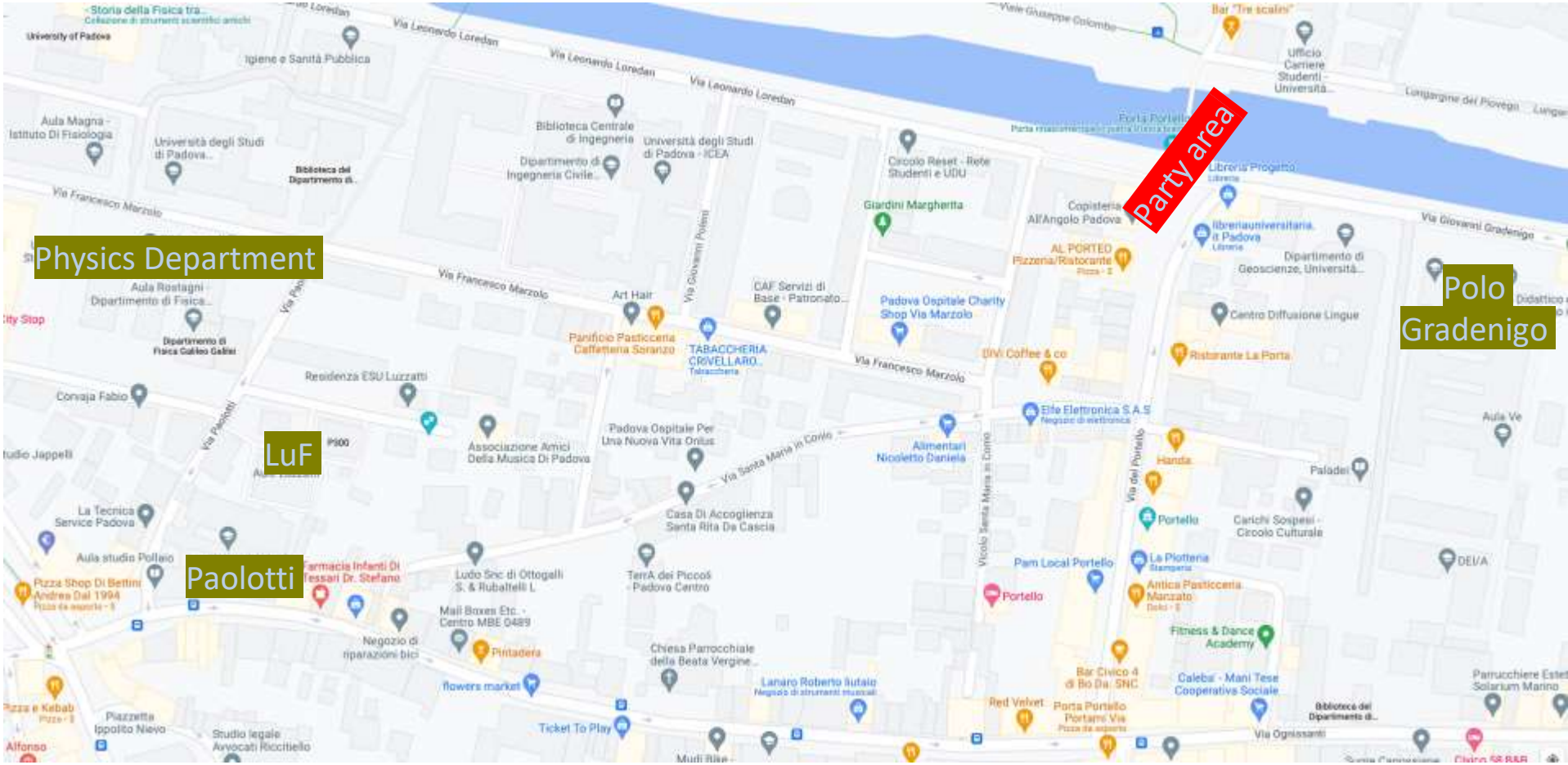
Timetable

- Schedule available on the AgendaStudenti platform
- Classes will be delivered in a dual modality, i.e. both in presence and online, for the first weeks to address issues related to the housing crisis and VISA issues
- Nota Bene: the attendance in presence to the practical lectures of the two labs is mandatory!

Timetable (first week)



Map


Physics Department
LuF
Paolotti
Party area
Polo Gradenigo
Via Francesco Marzolo
Via Leonardo Loredan
Via Giovanni Polleni
Via Paolotti
Via Santa Maria in Conio
Via del Portello
Via Giovanni Gradenigo
Igiene e Sanità Pubblica
Biblioteca Centrale di Ingegneria
Università degli Studi di Padova - ICEA
Dipartimento di Ingegneria Civile...
Circolo Reset - Rete Studenti e UDU
Giardini Margherita
Aula Magna - Istituto Di Fisiologia
Università degli Studi di Padova...
Aula Rostagni Dipartimento di Fisica...
Dipartimento di Fisica Galileo Galilei
Art Hair
Panificio Pasticceria Caffetteria Soranzo
TABACCHERIA CRIVELLARO...
CAF Servizi di Base - Patronato...
Padova Ospitale Charity Shop Via Marzolo
Copisteria All'Angolo Padova
AL PORTEO Pizzeria/Ristorante
Libreria Progetto
libreriauniversitaria.it Padova
Dipartimento di Geoscienze, Università...
Centro Diffusione Lingue
Ristorante La Porta
DIVI Coffee & co
Residenza ESU Luzzatti
Convaja Fabio
Associazione Amici Della Musica Di Padova
Padova Ospitale Per Una Nuova Vita Onlus
Casa Di Accoglienza Santa Rita Da Cascia
Alimentari Nicoletto Daniela
Elfe Elettronica S.A.S.
Handa
Portello
Carichi Sospesi - Circolo Culturale
Paladei
La Tecnica Service Padova
Aula studio Pollaio
Pizza Shop Di Bettini Andrea Dal 1994
Farmacia Infanti Dr. Stefano
Ludo Snc di Ottogalli S. & Rubaltelli L.
TerrA dei Piccoli - Padova Centro
Pam Local Portello
La Piotteria
Antica Pasticceria Manzato
DEI/A
Mail Boxes Etc. - Centro MBE 0489
Negozio di riparazioni bici
Printadera
flowers market
Chiesa Parrocchiale della Beata Vergine...
Lanaro Roberto Rutalo
Fitness & Dance Academy
Bar Civico 4 di Bo Da. SNC
Red Velvet
Porta Portello Portami Via
Caleba' - Mani Tese Cooperativa Sociale
Parrucchiere Estet Solarium Marino
Piazzetta Ippolito Nievo
Studio legale Avvocati Riccitiello
Ticket To Play
Via Ognissanti

Join Telegram group(s)



Physics of Data, cohort 26-27

Join Telegram group(s)



Physics of Data